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4. When magnesium ribbon is added to excess hydrochloric acid, HCl, magnesium chloride and hydrogen gas are produced.

magnesium + hydrochloric acid $\longrightarrow$ magnesium chloride + hydrogen

(a) Suggest **two** things that you would expect to **see** during this reaction. [2]

1. ....

2. ....

(b) Write the **symbol** equation for the reaction that takes place. [3]

.... + ... $\longrightarrow$ .... + ...

(c) Amelia recorded the volume of gas produced for 7 minutes for this reaction. She carried out the reaction at room temperature, 20 °C. The results she collected are shown in the table.

Time (minutes)	Volume of hydrogen produced (cm^3)
0	0
1	15
2	28
3	32
4	45
5	49
6	50
7	50

(i) Plot a graph of her results **on the grid opposite**. Draw a suitable line and label it **A**. [3]

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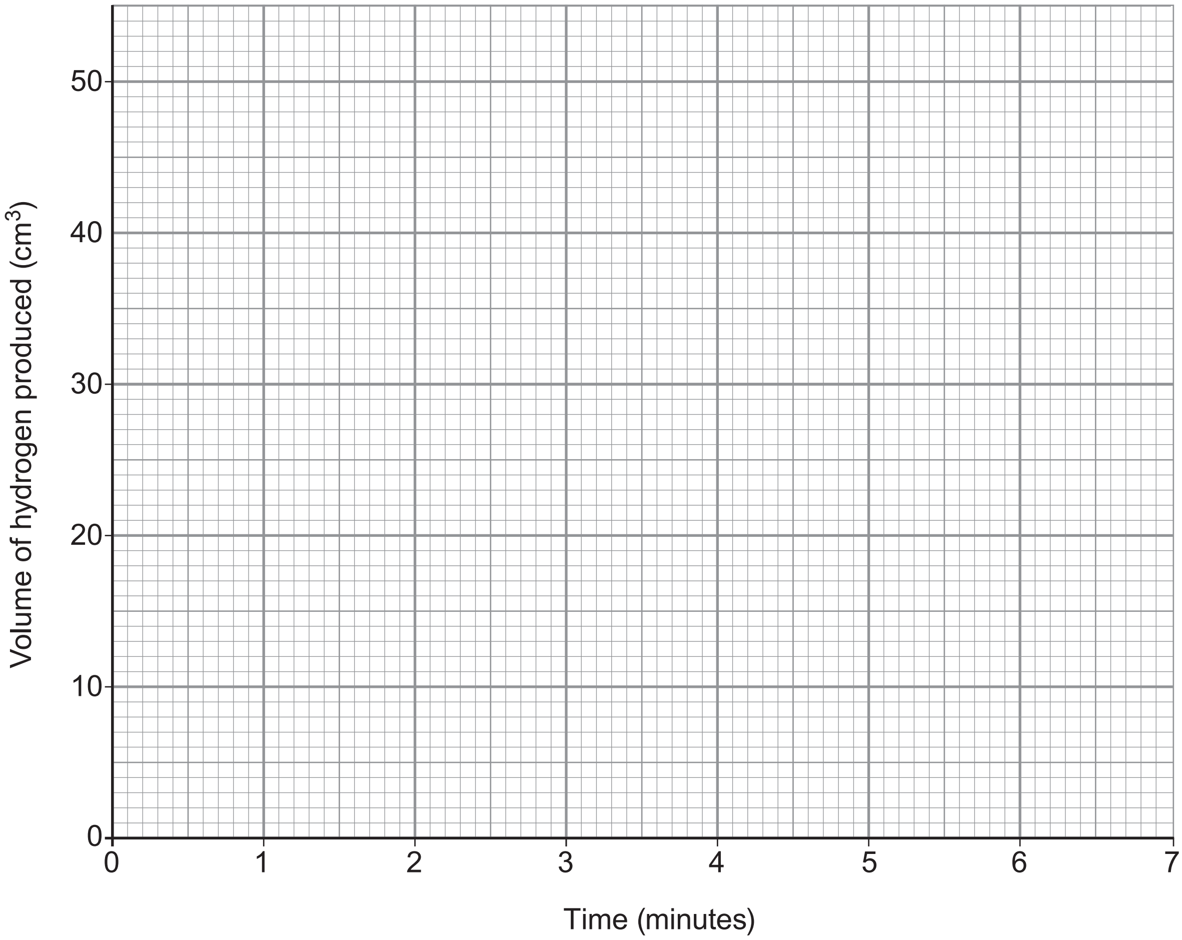

(ii) Use the graph to find:

I. the volume of gas produced in 90 seconds, ... [1]

II. the time taken for the reaction to end. .... [1]

(iii) **On the same grid**, sketch the graph that would be obtained if Amelia repeated the same experiment at 40 °C. Label it **B**. [2]

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